

Study

Anki flashcards on the Xteink X4 Pro. FSRS scheduling, your own decks, your own fonts. Reviews sync back to Anki.

The daily loop

1. Put a deck on the card

Convert from your live Anki collection, copy to the SD card.

2. Study

Apps > STUDY. Tap to reveal, tap a button to grade.

3. Sync back

One command applies your reviews and pushes to AnkiWeb.

Commands

Run from `firmware-next/`. **Close Anki first** for anything that writes.

Convert a deck (also picks up newly optimised FSRS parameters)

```
python3 tools_local/study/anki_to_deck.py \
~/Library/Application\ Support/Anki2/User\ 1/collection.anki2 \
--deck 'Mandarin: Vocabulary::Current' --name Mandarin \
--out /Volumes/SDCARD/study/mandarin
```

Sync reviews back (backs up first: idempotent. safe to re-run)

```
python3 tools_local/study/deck_to_anki.py \
/Volumes/SDCARD/study/mandarin \
~/Library/Application\ Support/Anki2/User\ 1/collection.anki2 --sync
```

Drop `--sync` to apply locally only. Add `--dry-run` to see what would change. AnkiWeb credentials come from `$ANKI_USERNAME / $ANKI_PASSWORD` or a prompt.

Build the fonts (once, or when the deck gains new characters)

```
.venv-study/bin/python tools_local/study/make_fonts.py \
--media ~/Library/Application\ Support/Anki2/User\ 1/collection.media \
--deck /Volumes/SDCARD/study/mandarin --out /Volumes/SDCARD/study/fonts
```

Check a deck before trusting it

```
.venv-study/bin/python tools_local/study/check_deck.py \
/Volumes/SDCARD/study/mandarin --fonts /Volumes/SDCARD/study/fonts
```

On the SD card

`/study/mandarin/` the deck (~70 KB)

`/study/fonts/` five CJK faces (~32 MB)

deck.dat card text, never written by the device.

cards.dat scheduling state.

revlog.dat your reviews, waiting to sync.

Retraining FSRS

There is no optimiser on the device and there should not be. Sync your reviews, then in Anki run **Deck options > FSRS > Optimize**. Re-run the convert command and the new weights travel to the device with the deck.

Good to know

Fonts change every card. Five faces, picked at random, so you never lock onto one letterform. Same five your Anki template uses.

Scheduling matches Anki. Same FSRS weights, same learning steps, read from your deck preset. Individual intervals differ by a few percent only because Anki randomises its own and this does not.

Reviews are never lost. Every answer is flushed to the card immediately, and sync is idempotent.

Back leaves at any point. Everything else is a tap.

Not there yet

Sentence images 290 of your 301 cards have one. The device shows text only, so those cards are missing a cue Anki gives you. The biggest gap.

Real hardware Everything so far is verified in the simulator through the real render path. Untested on the device itself.

One deck at a time The app reads `/study/mandarin`. Switching decks means reconverting.

Leeches Anki suspends a card after 8 lapses; you have 3. Here they keep coming back.

No editing No adding, editing, undo, or custom study. Do those in Anki.

No audio The X4 Pro has no speaker. Permanent.

Full detail: [docs/study-deck-format.md](#) · scheduler notes in `src/apps_local/study/StudyFsrs.h`